

SEEDLINE · CALCULATION NOTE

SeedLine sizing and first-principles checks

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (drive and spacing, metering, plates, mass, push force, slip, structure, work rate, depth, uniformity, cost) against every requirement

SeedLine sizing and first-principles checks

On paper the seeder works as a ground-driven single-row planter: the drive covers 22 to 589 mm of in-row spacing, slip from the metering load is under 2 %, the parts are lightly stressed and the work rate is about 0.145 ha/h. Ten of the eighteen requirements are met and two cannot be verified at TRL 3. Four are **at risk**: R1 (cell speed at 3 km/h is 0.31 m/s against an assumed 0.30 m/s limit), R2 (spacing CV about 31 % with assumed miss, double and scatter rates), R8 (depth holds ± 10 mm only for wheel-path bumps of about ± 15 mm) and R10 (136 N push in the design case, but 178 N at best on a heavy, loose seedbed). Two are **not met**: R11 (15.6 kg with the marker kit against 15 kg) and R17 (base seeder parts about \$214 against about \$200).

Every number in this note is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the key dimensions, sprocket geometry, cell sizes, hopper volume and handle geometry from the parametric model `cad/src/model.py`, and the prices from `bom/bom.csv`, so the model, the drawing SDL-DWG-001, the BOM and this note agree. All values are first-principles estimates; nothing here is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design unless marked as decided.

Input	Value	Basis
Architecture	Vertical cell plate, #35 chain, front drive wheel, rear press wheel, bolted frame, marker arm	Decided, SDL-DDR-001 items 1, 3, 4, 7 and 8
Design case	Maize at 250 mm on 0.75 m rows, 50 mm deep, 3 km/h (0.83 m/s)	SDL-REQ-001
Drive wheel	300 mm over the lug tips, 45 mm wide, 10 mm lugs; travel 942.5 mm per turn before slip	Model parameter
Typical slip for spacing tables	5 %	Allowance; the calculated skid (section 6) is lower
Cell fill limit	0.30 m/s cell speed	Assumed working limit for a gravity-filled cell plate; to be checked by bench test at TRL 4
Cell size	Length $1.15 L + 0.5$ mm, depth $1.05 W + 0.3$ mm, plate thickness the larger of 6 mm and $1.2 T$ (L, W, T: seed length, width, thickness)	Rule used by <code>make_plate</code> in the model
Seed dimensions	Maize $12 \times 8 \times 5$ mm, sorghum $4.5 \times 4 \times 3$, groundnut kernel $15 \times 9 \times 8$, common bean $13 \times 8 \times 6$, spinach $3.5 \times 3 \times 2.5$, pelleted carrot 3.3 mm, raw carrot $3 \times 1.2 \times 0.6$	Typical values; varieties differ
Print	± 0.2 mm on a 0.4 mm nozzle, tolerance limited to 10 % of the smallest cell dimension; 1.6 mm of skins, 1.2 mm walls, 25 % infill; 6 mm ³ /s average rate; PETG 1.27 g/cm ³ at \$25/kg	Common desktop FDM printer
Meter torque	1.0 N·m at the plate shaft (design value)	First-principles estimate is 0.20 N·m (section 3); the design value is five times higher
Soil	Cone index 300, 200 and 100 kPa for good, design and heavy (loose, cloddy) seedbeds; opener specific resistance 20, 30 and 40 kPa; covering chains and press 10, 15 and 20 N	Assumed; no soil data from a partner region yet
Rolling resistance	Brixius motion resistance, $MR/W = 1/Bn + 0.04$, with $Bn = CI b d / W / (1 + 3 b/d)$ for a rigid wheel	Brixius (1987), an empirical model fitted to larger tires; used here outside its calibrated range
Opener draft	Specific resistance $\times 2.5 \times 18$ mm runner width $\times$ depth	Narrow-tool failure zone taken as 2.5 tool widths
Field work	50 m rows, 15 s per turn, 90 s per refill, 85 % working time	Assumed for a small plot and one operator
Structure	S235 class steel, yield 235 MPa; dynamic factor 3 for drops and stones; 100 N side load at the grip	Assumed
Uniformity model	5 % misses, 4 % doubles (second seed 0 to 30 mm behind), 15 mm drop scatter, slip 5 ± 1.5 %	The miss and double rates are the TRL 2 estimates; not calculable from first principles

■ 2. Drive, spacing and range (R3, R4)

The spacing is $s = \pi D / (n i) / (1 - \sigma)$, with $\pi D = 942.5$ mm, n plate cells, ratio $i = \text{wheel teeth} / 15$ and travel reduction σ . The three wheel sprockets give $i = 0.8, 1.0$ and 1.2 (pitch diameters 36.80, 45.81 and 54.85 mm; plate sprocket 15 T).

Table 2. Example plates. Nominal spacing, and with the 5 % slip allowance.

Crop	Target (mm)	Cells	Ratio	Nominal (mm)	With 5 % slip (mm)
Maize (design case)	250	4	1.0	235.6	248.0
Sorghum, groundnut	150	6	1.0	157.1	165.3
Common bean	100	9	1.0	104.7	110.2
Spinach	50	16	1.2	49.1	51.7
Carrot, lettuce (pelleted)	25	30	1.2	26.2	27.6
Skip-cell plate	400	2	1.2	392.7	413.4
Skip-cell plate	450	2	1.0	471.2	496.0
Skip-cell plate	600	2	0.8	589.0	620.1
Shortest		36	1.2	21.8	23.0

With skip-cell plates (one or two cells; decided, SDL-DDR-001 item 5) the nominal range is **21.8 to 589 mm**, and 1,178 mm with a single cell, so **R3 (25 to 400 mm) is met**. The settings are discrete: choosing the best plate and ratio for every target from 25 to 400 mm, the worst error is 11.1 % (a 372 mm target falls between 331 mm and 413 mm). The spacing card should list the nearest setting for each crop.

Chain. The sprocket centers are 286.2 mm apart. A #35 chain needs 73.61, 75.10 and 76.61 pitches for the 12, 15 and 18 T wheel sprockets, so the chain is set to 74, 76 or 77 links (the 77-link length needs one offset link). The largest slack left over is 8.6 mm (15 T, 76 links), which a spring idler takes up by deflecting the slack strand by about 35 mm. The idler in `bom/bom.csv` has 45 mm of travel.

■ 3. Metering speed and cell fill (R1)

At 3 km/h the wheel turns at 53.1 rpm. For the maize plate the seed centers sit on a 111.3 mm circle, so:

Table 3. Plate and cell speeds at 3 km/h.

Ratio	Plate speed (rpm)	Cell speed (m/s)	Walking speed for 0.30 m/s (km/h)
0.8	42.4	0.247	3.64
1.0	53.1	0.309	2.91
1.2	63.7	0.371	2.43

The design case runs at 0.309 m/s, just above the assumed 0.30 m/s limit, so **R1 is at risk**: the operator should walk at about 2.9 km/h, or 2.4 km/h with a 1.2 ratio plate. A simple kinematic indicator (a still seed must fall the cell depth while the open cell passes it) gives 0.20 m/s for the maize cell; seeds in the pool are dragged along by the

plate, so real plates fill at higher speeds, but the indicator confirms that cell speed, not the drive, sets the walking speed. The maize plate passes 3.5 cells per second; the 36-cell plate at 1.2 passes 38.2 cells per second, which is demanding for the brush. Misses and multiples depend on seed shape, brush setting and cell speed and cannot be calculated on paper; they need the bench test that TRL 4 would bring.

Meter torque. Friction of the seed pool on the lower quarter of both plate faces (Janssen-type wall pressure under a 0.2 m seed head) is about 0.051 N·m, the brush about 0.048 N·m and the bearings and chain about 0.10 N·m, a total of **0.20 N·m**. The calculations below use 1.0 N·m, five times this estimate, to cover seed jamming at the brush.

A smaller plate would lower the cell speed in proportion to the cell circle: see the review note for this as an option.

■ 4. Plates per crop (R5, R6)

Table 4. Cell sizes, print tolerance, mass and print time for each starter plate.

Plate	Cell length x depth (mm)	Plate thickness (mm)	Cells (maximum)	Tolerance / smallest cell dimension	Mass (g PETG)	Print time (h)
Maize	14.3 x 8.7	6	4 (21)	2.3 %	40.0	1.46
Sorghum	5.7 x 4.5	6	6 (47)	4.4 %	41.1	1.50
Groundnut kernel	17.8 x 9.8	10	6 (17)	2.1 %	53.8	1.96
Common bean	15.4 x 8.7	8	9 (20)	2.3 %	45.8	1.67
Spinach	4.5 x 3.5	6	16 (56)	5.8 %	41.1	1.50
Carrot, pelleted	4.3 x 3.8	6	30 (58)	5.3 %	41.1	1.50
Carrot, raw	3.9 x 1.6	6	30 (62)	12.8 %, too fine	41.2	1.50

Raw seed from 3.5 mm (spinach) to 15 mm (groundnut) fits the tolerance rule. Raw carrot seed needs a 1.6 mm deep cell, where ± 0.2 mm is 12.8 % of the cell, so 1 to 2 mm seed must be sown pelleted (about 3.3 mm pellets), as decided (SDL-DDR-001 item 6). With that redefinition **R5 is met** on paper. Every starter plate prints in 2 h or less, but the 10 mm groundnut plate takes 1.96 h, so **R6 is met with almost no margin** for thick plates; the ± 0.2 mm cell accuracy cannot be verified at TRL 3. The TRL 2 estimate of about 25 g and 1 h per plate was low; the model gives 40 to 54 g and 1.5 to 2.0 h.

■ 5. Mass and center of mass (R11)

Table 5. Mass by group.

Group	Mass (kg)	Notes
Wheels and axles (items 1, 11)	3.94	2.2 kg drive wheel, 1.2 kg press wheel, two 16 mm axles
Drive and metering (items 2 to 8)	2.00	Hopper from the model volume, 3 mm PETG walls
Opener and covering (items 9, 10)	1.59	
Frame and handle (items 12, 13)	6.61	Frame 3.63 kg (1.65 m of 25 x 25 x 1.5 tube, drops, posts, brackets); handle 2.99 kg (two 978 mm side tubes of 25 x 1.5 round tube, grip, brace, sleeves)
Hardware (item 15)	0.45	
Base seeder	14.59	
Row marker kit (item 14)	1.00	
With marker kit	15.59	

The base seeder meets 15 kg with 0.4 kg to spare, but with the marker kit fitted it is 15.6 kg, so **R11 is not met**. The TRL 2 figure (about 14.1 kg with the marker) left out the extra tube length and brackets of the bolted frame and underestimated the handle. The center of mass of the base seeder is 383 mm behind the drive axle and 260 mm above the ground; a full hopper adds 1.73 kg of maize. The rails were sized down from 2 mm to 1.5 mm wall because the stresses are low (section 7); a lighter handle is the next place to save mass.

■ 6. Push force and slip (R10, R4)

The push is found from the planar statics of the seeder on its two wheels, with the operator's force at the grip (950 mm high, 647 mm behind the press wheel) at an angle ϕ below the horizontal. Rolling resistance acts at ground level and the opener draft 25 mm below it. For each case the script finds the smallest horizontal push that balances the resistance with both wheels on the ground.

Table 6. Push force (horizontal component) with a full hopper, 160 N total weight.

Seedbed	Along the handle (50°)	Lowest push, at angle	Angles that balance	Rolling / opener / chains (N)	Wheel loads, drive / press (N)
Good	83 N	71 N at 16°	7 to 50°	28 / 45 / 10	85 / 174
Design	136 N	109 N at 28°	20 to 50°	54 / 68 / 15	103 / 220
Heavy, loose	no balance	178 N at 35°	29 to 47°	68 / 90 / 20	214 / 71 (at 35°)

Three findings follow. First, a purely horizontal push has no balance in any case: at grip height it pitches the seeder forward and lifts the press wheel, so the TRL 2 comparison of a horizontal push against a push along the handle does not apply. Second, in the design case the push along the handle is 136 N, inside the 150 N target with a 9 % margin that rests on assumed soil values. Third, on a heavy, loose seedbed a push along the handle drives the press

wheel into the soil faster than it moves the seeder (no balance), and the best push is 178 N at 35°. **R10 is at risk.** The equivalent rolling coefficients are 0.107, 0.167 and 0.239, inside the 0.2 to 0.3 range assumed at TRL 2 only for the heavy case.

Slip. The drive wheel must supply the meter torque through the chain: 6.7 N at the rim at a 1.0 ratio and 8.0 N at 1.2. Against 85 to 214 N of wheel load, the Brixius traction relation gives a skid of **1.2 to 1.7 %**, well inside the 8 % of R4. The larger uncertainty is the rolling radius: if the lugs sink fully, the wheel rolls on its rim and travels up to 6.7 % less per turn, which makes the spacing shorter, not longer. The two effects partly cancel. **R4 is met on paper;** the spacing card should be calibrated by counting wheel turns over a measured 10 m on the user's own seedbed.

■ 7. Structure

Table 7. Stress checks.

Part	Load case	Result
Side rails, 25 x 25 x 1.5 tube	Heavy case wheel and push loads, factor 3, plus opener clamp moment	70 MPa, factor 3.4 on yield
Handle side tube, 25 x 1.5 round	100 N side load at the grip	80 MPa, factor 2.9
Press axle, 16 mm	Heavy case load, factor 3	19 MPa
#35 chain	1.2 N·m on the 15 T plate sprocket	52 N against about 7.8 kN minimum tensile strength (typical catalog value)
Plate hub D-flat	1.2 N·m in PETG	5.7 MPa bearing stress

The frame and drive are lightly stressed. Durability (R15) depends on wear of the plates, brush, chain and opener, which cannot be calculated at TRL 3.

■ 8. Work rate and hopper (R9, R12)

The theoretical capacity on 0.75 m rows at 3 km/h is 0.225 ha/h. With 50 m rows, 15 s turns, 90 s refills and 85 % working time the field efficiency is 65 %, so the field capacity is **0.145 ha/h (6.9 h/ha)**, which meets R9 and compares with about 56 h/ha for hoe planting (SDL-PRB-001). If the walking speed is cut to 2.9 km/h for R1, the capacity falls in proportion to about 0.14 ha/h. On 0.3 m vegetable rows the capacity is 0.058 ha/h.

The hopper cavity in the model is **2.40 L** (R12 met). It holds 1.73 kg of maize, about 5,754 seeds, enough for 1,438 m of row or 0.108 ha per fill. Maize at 0.75 x 0.25 m needs 53,333 seeds per hectare.

■ 9. Depth control (R8)

The depth bracket gives 10 to 60 mm in 10 mm steps (six holes in the shank). The opener point is 182 mm behind the drive axle, so the frame, riding on its two wheels, raises the opener by 0.69 of any rise under the drive wheel and 0.31 of any rise under the press wheel. Holding ± 10 mm therefore needs the wheel path to stay within about ± 15 mm under the drive wheel or ± 32 mm under the press wheel. A 50 mm clod under the drive wheel lifts the opener by 34 mm. **R8 is at risk** on rough, hand-tilled seedbeds; a spring-loaded opener or a depth-gauge shoe would decouple the opener from the wheels, but that is a design change for review.

■ 10. Spacing uniformity (R2)

A Monte Carlo run of 20,000 cells for the maize plate, with the assumed rates in Table 1, gives a coefficient of variation of **30.7 %** for all plant-to-plant spacings, against a target of 30 % or less. The precision of the single spacings (those between 0.5 and 1.5 times the mean) is 8.9 %, with a miss index of 4.8 % and a multiple index of 3.9 %. The all-spacings CV is driven almost entirely by the misses and doubles, not by drop scatter or slip. **R2 is at risk**; it hinges on miss and double rates that only a bench test can give.

■ 11. Cost (R16, R17)

All 15 BOM lines are priced. The base seeder (items 1 to 13 and 15) costs **\$213.50** and the marker kit \$24.00, a total of **\$237.50** against the \$400 budget in `project.yaml` (R16 met). The base cost exceeds the about \$200 target of R17 by about 7 %, so **R17 is not met**. The main causes are the four #35 sprockets and idler for the decided ratio change (\$47) and the extra brackets and bolts of the bolted frame.

■ 12. Results against requirements

Table 8. Every requirement in SDL-REQ-001 v0.3, not met items first.

ID	Requirement	Value (this note)	Target	Status
R11	Mass and handling	14.6 kg base; 15.6 kg with the marker kit	15 kg or less	Not met
R17	Replication cost	\$214 base seeder	About \$200 or less	Not met
R1	Single-seed placement	Cell speed 0.31 m/s at 3 km/h against an assumed 0.30 m/s limit	Misses and multiples 5 % or less each	At risk
R2	Spacing uniformity	CV 31 % of all spacings; 9 % for singles (assumed rates)	CV 30 % or less	At risk
R8	Sowing depth	10 to 60 mm in 10 mm steps; ± 10 mm only for wheel-path bumps of about ± 15 mm	± 10 mm on a tilled seedbed	At risk
R10	Push effort	136 N along the handle (design case); 178 N at best on a heavy, loose seedbed	150 N or less	At risk
R7	Crop change	Side door and hand knob in the model	2 min or less, no tools	Not verifiable at TRL 3
R15	Durability	Stresses low (section 7); wear life unknown	5 seasons; plates 1 season	Not verifiable at TRL 3
R3	Spacing range	22 to 589 mm nominal; every 25 to 400 mm target within 11.1 %	25 to 400 mm	Met
R4	Spacing follows travel	Skid 1.2 to 1.7 %; rolling radius uncertain by up to 6.7 %	Slip 8 % or less	Met
R5	Seed size range	3.5 to 15 mm raw seed; 1 to 2 mm seed as pellets of about 3.3 mm	1.5 to 15 mm, seed under 2 mm pelleted	Met
R6	Printable plates	1.46 h and 40 g (maize); up to 1.96 h (groundnut); accuracy not verifiable	2 h or less	Met, no margin for thick plates
R9	Work rate	0.145 ha/h (6.9 h/ha)	0.1 ha/h or more	Met
R12	Hopper	2.40 L; 1,438 m of maize row per fill	2 L or more	Met
R13	Row-spacing kit	Marker reach 200 to 900 mm, set at 750 mm in the model	200 to 900 mm, ± 25 mm	Met (design review)
R14	Guarding	Band guard with an outboard face plate over both sprockets and the idler	Nip points covered	Met (design review)
R16	Prototype cost	\$238	\$400 or less	Met
R18	Local build	Bolted 25 mm square tube, 6 mm plate, #35 chain	Common sections, drill and bolts	Met (design review)

Summary: 10 met, 4 at risk, 2 not met, 2 not verifiable at TRL 3.

■ 13. Corrections to the TRL 2 figures

Quantity	TRL 2 (SDL-PRC-001 v0.2)	TRL 3 (this note)
Spacing range	About 22 to 393 mm; R3 not met	21.8 to 589 mm with skip-cell plates; R3 met
Plate mass and print time	About 25 g, about 1 h	40 to 54 g, 1.5 to 2.0 h
Hopper	About 2.1 L, about 1,250 m per fill	2.40 L, 1,438 m per fill
Mass	13.1 kg base, 14.1 kg with marker	14.6 kg base, 15.6 kg with marker
Push force	About 100 N good, about 220 N heavy	83 N good, 136 N design, 178 N heavy at best
Slip	3 to 8 %	1.2 to 1.7 % skid, plus up to 6.7 % rolling radius uncertainty
Work rate	About 0.14 ha/h	0.145 ha/h
Parts cost	\$199 base, \$223 with marker	\$214 base, \$238 with marker
Meter torque	0.5 to 1.0 N·m	0.20 N·m estimated; 1.0 N·m used

Safety: These figures are paper estimates. The chain drive, opener point and wheel lugs remain hazards whatever the numbers say; the guard must be fitted in use and plate changes done with the wheel held still (SDL-PRC-001, Safety). Treated seed is toxic; follow the seed label.